

B2 position and removal of B3

Preferred study arrangement: remove B3 and move B2 to $y = 161.375$ inches north of the south garage wall: 13 ft 5-3/8 in. This is 23-5/8 inches south of its previous $y = 185$ -inch location. Use the full T-N lower chord as the north floor support. The resulting two joist spans are each 91.625 inches (7 ft 7-5/8 in).

B3 was a floor-level beam in the previous model. Removing it is not deletion of the floor support: its job is transferred to the T-N truss lower chord through joist seats/connections that still need detailing. This comparison gives no new load credit to the existing north garage wall.

	One B2	Two beams
Positions north of south wall	161.375 in	130.833 and 191.917 in
Candidate interior rails	W8×31	W8×24 each
Longest joist span	91.625 in	61.083 in
Primary steel + 10% connection allowance	7,286 lb	7,641 lb
Maximum gravity screen ratio	0.809	0.804
Maximum global service movement	0.747 in	0.743 in

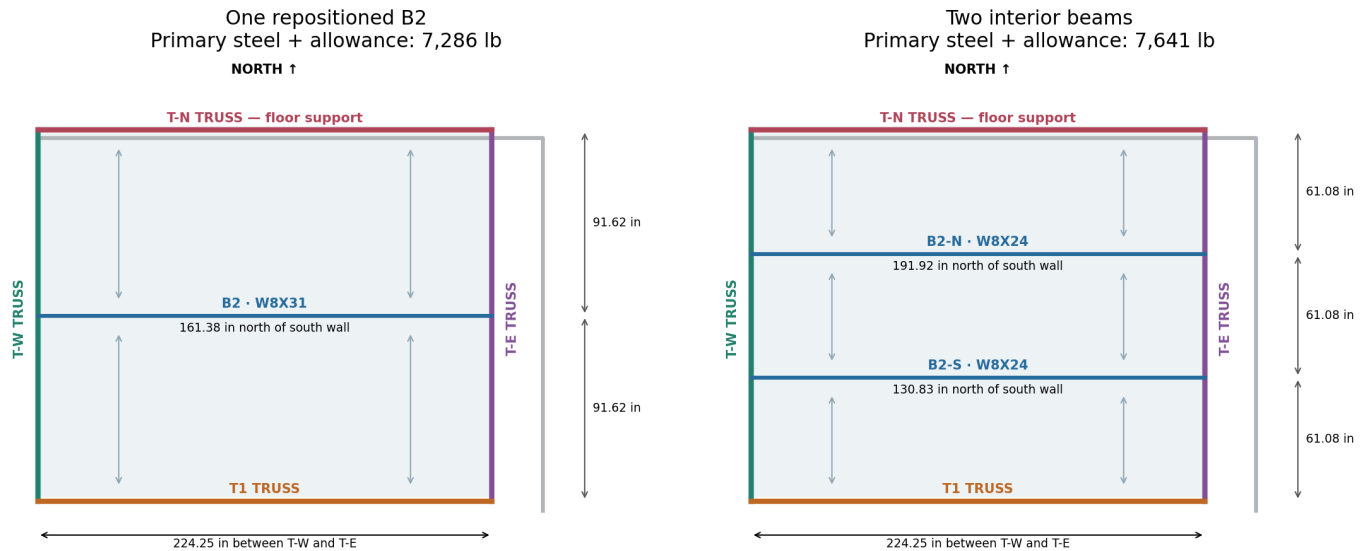
The single-beam option is about 355 lb lighter in primary steel. The two-beam option shortens the longest joist span by one third, which may allow shallower or lighter secondary floor framing. The 9 psf floor dead-load allowance is held constant in this study; secondary joist sizing/cost has not been optimized, so the whole-floor weight advantage is not settled.

Several single-beam positions have the same minimum tested primary weight. The midpoint is preferred because it minimizes the longest joist span between fixed T1 and T-N supports, not because the steel search found a unique mathematical optimum. This is a finite position/section comparison with selected companion-member upsizing.

Both arrangements retain the preceding T1 option: W6×16 lower chord, W6×20 upper chord, HSS 2×2×1/8 verticals and HSS 2½×2½×1/8 end diagonals. The 1,000 lb lifted trolley load and provisional dynamic/equipment allowances are retained. All columns and the west O3 transfer beam remain at their existing coordinates.

Floor framing comparison

B3 removed • north floor joists bear on T-N's lower chord



Floor-support plan only. Existing columns remain fixed; O3 remains at $y = 185$ in.
Conditional gravity comparison: joint/seat details, bracing and trolley local checks remain unresolved.

Arrows inside the deck indicate north-south floor-joist spanning direction. Dimension chains show support spans; they are not W-beam spans. Both W beams still span 224.25 inches east-west between T-W and T-E. The floor remains open south of $y = 72$ inches. The deck extends to $y = 255$ inches, giving a 2-inch north overhang past T-N at $y = 253$ inches.

B2 no longer needs to line up with W3: it transfers through T-W, which connects to the unchanged O3/W3 support line at $y = 185$ inches. That indirect load path is included in the connected model; its joints and restraint must be designed. No extra columns or garage-floor footings have been added by this comparison.

The drawing shows a possible floor-support arrangement, not approved connection or fabrication details. The assumed T-N bearing detail must clear the north door, and any rail/seat connections must preserve trolley passage.

Position and section sweep

B2 position(s), in	Rail section(s)	Steel lb	Ratio	Movement in	Screen
130.00	W8X31	7,286	0.786	0.753	Exceeds
145.00	W8X31	7,286	0.780	0.744	Meets
161.38	W8X31	7,286	0.809	0.747	Meets
175.00	W8X31	7,286	0.870	0.745	Meets
185.00	W8X31	7,286	0.888	0.740	Meets
200.00	W8X31	7,286	0.850	0.738	Meets
120.00, 185.00	W8X24/W8X31	7,785	0.800	0.712	Meets
130.83, 191.92	W8X24/W8X24	7,641	0.804	0.743	Meets
140.00, 205.00	W8X31/W8X24	7,779	0.891	0.722	Meets

The limits are the inherited 0.90 combined-force screening metric and 0.75-inch maximum absolute global service movement. “Meets” means only those conditional gravity screens. The 130-inch trial exceeds the arbitrary movement threshold by about 0.003 inch; this does not demonstrate that such a layout is physically impossible. Candidate families and connection assumptions materially affect the result.

For each layout, the Python model regenerates member intersections and the floor tributary-load distribution. It removes B3 and places the north floor reactions on T-N, checks the selected frame, upsizes failing companion families where possible, and tests W-rail section combinations. Other truss members are held fixed except required upsizing; a simultaneous global reoptimization could change the ranking.

There are six one-beam positions and three two-beam layouts. Rail candidates are actual catalogue W sections at or below 8 inches actual depth. Trolley loads are sampled at nine locations on each rail and on T1, one operating trolley at a time. The whole connected frame is solved for each option; no truss is assigned the entire building load independently.

What happens to T-N?

Case	T-N lower chord	Its controlling gravity ratio
One centered B2	HSS4x4x0.188	0.544
Two equal-spaced beams	HSS4x4x0.188	0.447

The T-N lower chord can serve as the north floor collector in this conditional model. Its own bending, axial force and connection behavior still matter: being part of a truss does not make local floor-bearing effects disappear. The truss shares load through its upper chord, web system and supporting columns.

The one-beam option has little room below the chosen 0.75-inch global movement target. That target is not a code-specific relative beam deflection or trolley alignment check. This result should guide layout selection, not final member purchase. Joist deflection, the separately discussed 1,000 lb concentrated floor item, trolley flange-local checks and floor vibration still need member-level design.

Unchanged limitations: first-order analysis; assumed rigid joints and fixed bases; assumed chord brace spacing; simplified W torsion without restrained warping; unverified roof collectors and diaphragm action; no completed site wind/seismic, connection, anchor or footing design. North floor seating into T-N and T-W/O3 transfer details are especially important for this revision.

Reproduction: run run_sweep.py with the FreeCAD Python environment or equivalent NumPy/SciPy runtime; the balanced single-beam case is generated with BEAM_YS=[161.375] and LAYOUT=single-161 via test_layout.py. The scripts depend on the preceding revision-4 selected-options.json. Full per-layout selections, member demands and column reactions are in results/. make_plans.py and report.py generate this comparison. The AISC/Atlas section data and load references are those documented in revision 3.